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1 Branch 0 Tags Go to file T Go to file Add file Code

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LICENSE	Initial commit	3 months ago
README.md	Update README.md	now
Untitled3.ipynb	Add files via upload	3 minutes ago

README License

edge-detection-opencv

Aim

To perform edge detection using Sobel, Roberts, Prewitt, Laplacian, and Canny edge detectors.

Software Required

- Anaconda – Python 3.7
- Jupyter Notebook / VS Code
- OpenCV (cv2)
- NumPy
- Matplotlib

⚙️ Algorithm

Step 1:

Import all the necessary modules for the program.

Step 2:

Load an image using `cv2.imread()` .

Step 3:

Convert the image to grayscale.

Step 4:

Apply **Sobel operator** using OpenCV to detect edges.

Step 5:

Apply **Prewitt operator** using custom kernels.

Step 6:

Apply **Roberts operator** using custom kernels.

Step 7:

Apply **Laplacian operator** using OpenCV.

Step 8:

Apply **Canny edge detector** using OpenCV.

Step 9:

Display all edge-detected images for comparison.

Developed By

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- **Register No:** 212225240009

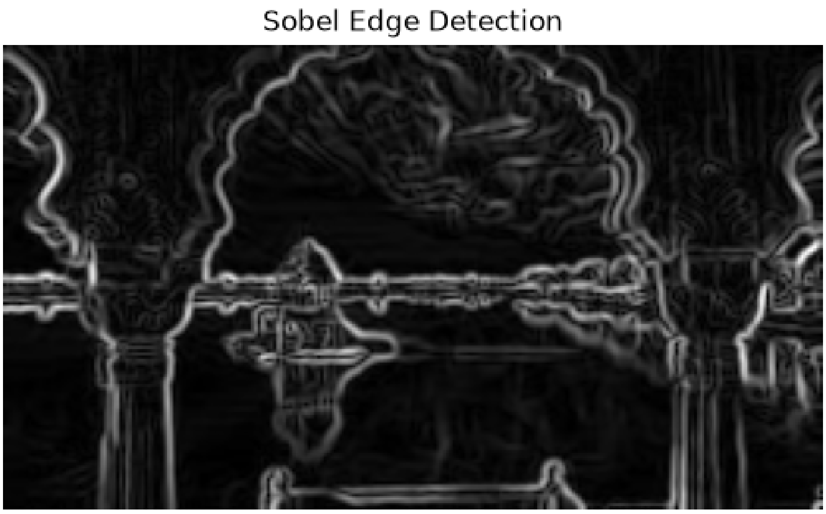
Output

Sobel Edge Detector

```
import cv2
import numpy as np
import matplotlib.pyplot as plt

image = cv2.imread('place.jpg')
gray_image = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)

plt.imshow(cv2.cvtColor(image, cv2.COLOR_BGR2RGB))
plt.title('Original Image')
plt.axis('off')
```

Prewitt Edge Detector

```
sobel_x = cv2.Sobel(gray_image, cv2.CV_64F, 1, 0, ksize=5)
sobel_y = cv2.Sobel(gray_image, cv2.CV_64F, 0, 1, ksize=5)
sobel_combined = cv2.magnitude(sobel_x, sobel_y)
plt.imshow(sobel_combined, cmap='gray')
plt.title('Sobel Edge Detection')
plt.axis('off')
```

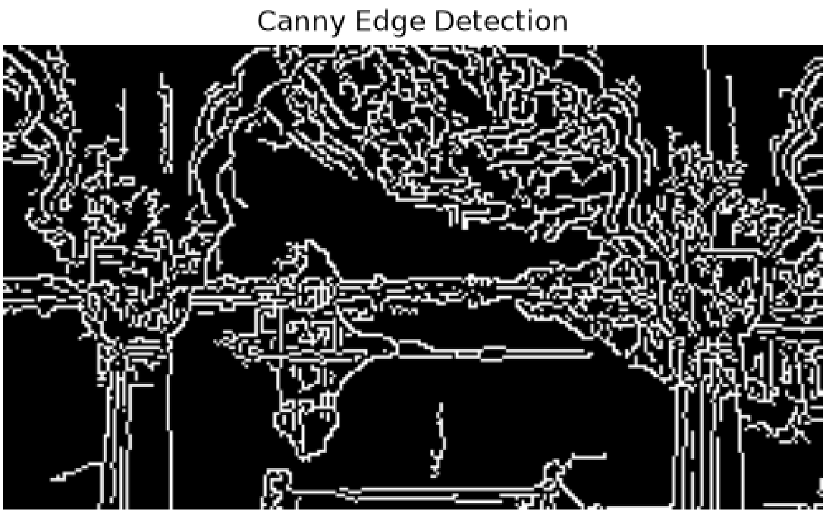


Roberts Edge Detector

```
laplacian = cv2.Laplacian(gray_image, cv2.CV_64F)
plt.imshow(laplacian, cmap='gray')
plt.title('Laplacian Edge Detection')
plt.axis('off')
```

Laplacian Edge Detector

```
canny_edges = cv2.Canny(gray_image, 50, 150)
plt.imshow(canny_edges, cmap='gray')
plt.title('Canny Edge Detection')
plt.axis('off'
```

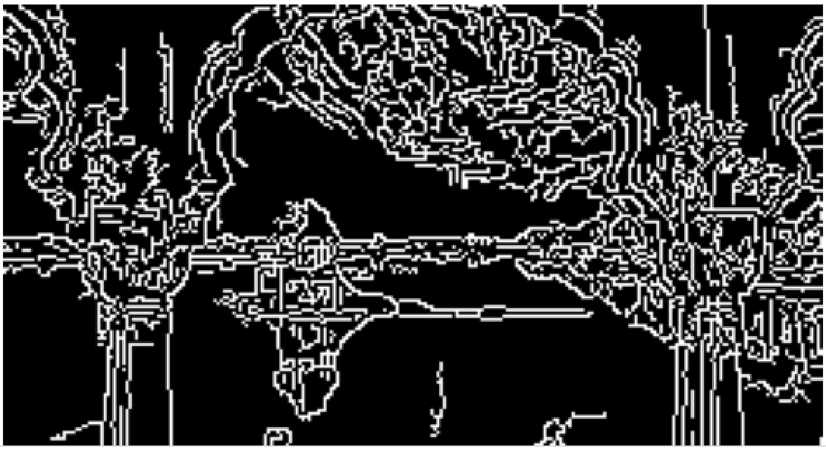


Canny Edge Detector

```
image = cv2.imread("place.jpg")

gray = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)
prewitt_x = np.array([[1, 0, -1],
                      [1, 0, -1],
                      [1, 0, -1]])

prewitt_y = np.array([[1, 1, 1],
                      [0, 0, 0],
                      [-1, -1, -1]])

prewitt_x_edge = cv2.filter2D(gray, -1, prewitt_x)
prewitt_y_edge = cv2.filter2D(gray, -1, prewitt_y)
prewitt = cv2.magnitude(prewitt_x_edge.astype(np.float32),
                        prewitt_y_edge.astype(np.float32))

plt.imshow(canny_edges, cmap='gray')
```

```
    plt.title('Prewitt Edge Detection')
    plt.axis('off')
```

Prewitt Edge Detection



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